

Q1. DIRECTIONS *for questions 1 to 3:* Select the correct alternative from the given choices.

If A is 25% as efficient as B and can complete a certain work taking 15 days more than the time taken by B, in how many days will both A and B together complete the work?

☒ a) 4 days ✓ **Your answer is correct**

☐ b) 5 days

☐ c) 7 days

☐ d) 10 days

Q2. DIRECTIONS *for questions 1 to 3:* Select the correct alternative from the given choices.

If $n > 5$, find the remainder when $7^n - 6n(3n - 2) - 7$ is divided by 216.

- ☐ a) **43**
- ☐ b) **1**
- ☐ c) **210**
- ☐ d) **Cannot be determined**

Q3. DIRECTIONS *for questions 1 to 3:* Select the correct alternative from the given choices.

There are two variants of detergents – A and B. Detergent A contains 75% soap and 25% bleach, while detergent B contains 85% soap and 15% bleach. In what ratio should the detergents A and B be mixed, so that the resultant mixture contains 79% soap in it?

☒ a) 2 : 3 ✖ Your answer is incorrect

☐ b) 3 : 4

☐ c) 5 : 4

☐ d) 3 : 2

Q4. DIRECTIONS *for question 4:* Type in your answer in the input box provided below the question.

If the first four terms of an arithmetic progression are p , $p + 2q$, $3p + q$ and 30 respectively, find the value of the 2016th term of the progression.

Q5. DIRECTIONS *for question 5:* Select the correct alternative from the given choices.

Two sellers, A and B, sell the same product but at a profit of 15% and 12% respectively. If the cost price of A is 85% of the cost price of B, find the ratio of the selling prices charged by A and B.

- ☐ a) 25 : 28
- ☐ b) 28 : 25
- ☒ c) 391 : 448 ✓ **Your answer is correct**
- ☐ d) 447 : 526

Q6. DIRECTIONS *for question 6:* Type in your answer in the input box provided below the question.

Two trains, of lengths 500 m and 300 m respectively, run on parallel tracks. When the trains run in the same direction, the faster train takes 80 seconds to cross the slower train. When they run in opposite directions, they take 20 seconds to cross each other. Find the speed (in km/hr) of the slower train.

Q7. DIRECTIONS *for question 7:* Select the correct alternative from the given choices.

The compound interest on a certain sum at a certain rate of interest for the 4th year is ₹2,420 and for the 5th year is ₹2,662. Find the compound interest for the 2nd year.

- ☐ a) ₹1811
- ☐ b) ₹2000
- ☐ c) ₹2100
- ☐ d) ₹2200

Q8. DIRECTIONS *for questions 8 and 9:* Type in your answer in the input box provided below the question.

The number of integral values k can take, where $2 \leq k \leq 30$, such that the product $(k - 1) \cdot (k - 2) \dots (3) \cdot (2) \cdot (1)$ is not divisible by k , is

Q9. DIRECTIONS *for questions 8 and 9:* Type in your answer in the input box provided below the question.

A binary operator $(*)$ is defined such that $(a * b) = a^b$, if $a \leq b$ and $= b^a$, if $b < a$. Find the value of $((2 * 3)^*3) + ((3 * 2)^*2)$.

Q10. DIRECTIONS *for questions 10 to 14:* Select the correct alternative from the given choices.

A boat started from a point in a river and traveled a certain distance upstream, after which it turned back and returned to its starting point. If the boat covered the round-trip journey in two hours and the net speed of the boat upstream was 25% of that downstream, how much more time (in minutes) did the boat take to travel upstream than what it took to travel downstream?

☐ a) 48

☐ b) 54

☐ c) 60

☐ d) 72

Q11. DIRECTIONS *for questions 10 to 14:* Select the correct alternative from the given choices.

If $0 < x < \frac{1}{2}$, which of the following is the greatest?

☐ a) x^{-2}

☐ b)

$x^{-\frac{3}{2}}$

☐ c) $\frac{1}{\sqrt{x}}$

☐ d) $\left(\frac{1}{x^{-2}}\right)^{-3}$

Q12. DIRECTIONS for questions 10 to 14: Select the correct alternative from the given choices.

Two positive real numbers, a and b , are expressed as the sum of m positive real numbers and n positive real numbers respectively as follows:

$$a = s_1 + s_2 + \dots + s_m$$

$$b = t_1 + t_2 + \dots + t_n$$

For any real number x , $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x and $\lceil x \rceil$ denotes the least integer greater than or equal to x .

If $\lfloor a \rfloor = \lfloor s_1 \rfloor + \lfloor s_2 \rfloor + \dots + \lfloor s_m \rfloor - 11$ and $\lceil b \rceil = \lceil t_1 \rceil + \lceil t_2 \rceil + \dots + \lceil t_n \rceil + 7$, what is the minimum possible value of $m + n$?

- ☐ a) 21
- ☐ b) 20
- ☐ c) 19
- ☐ d) 18

Q13. DIRECTIONS *for questions 10 to 14:* Select the correct alternative from the given choices.

Manohar bought 236 pens at a certain price. If he then sold all the pens at a fixed selling price each, such that he was able to recover his investment from the revenue obtained by selling exactly 200 pens, find his profit percentage.

- ☐ a) 18%
- ☐ b) 24%
- ☒ c) 36% **✗ Your answer is incorrect**
- ☐ d) 32%

Q14. DIRECTIONS *for questions 10 to 14:* Select the correct alternative from the given choices.

Find the maximum value of x such that $\sqrt{x} \geq 3x$.

☐ a) $\frac{1}{\sqrt{3}}$

☐ b) $\frac{1}{3}$

☐ c) $\frac{1}{9}$

☐ d) $\frac{1}{6}$

Q15. DIRECTIONS *for question 15:* Type in your answer in the input box provided below the question.

Three men and four women working together can do a piece of work in four days, whereas four men and two women working together require six days to complete the same work. In how many days can nine men and two women working together finish the work?

Q16. DIRECTIONS *for question 16:* Given below is a question statement followed by two statements, I and II, each giving some information. Study the sufficiency of the data given in each of the statements to answer the question and input your answer using the following instructions:

Input your answer as

1. if the question can be answered by using statement I alone, but cannot be answered by using statement II alone.
2. if the question can be answered by using statement II alone, but cannot be answered by using statement I alone.
3. if the question can be answered by using either statement alone.
4. if the question can be answered by using both the statements together, but cannot be answered by using either statement alone.
5. if the question cannot be answered even by using both the statements together.

If a , b and c are single-digit natural numbers, such that $a < b < c$, and the roots of the equation $ax^2 + 2bx + c = 0$ are equal, then find a .

I. a is not prime.

II. a is not composite.

Q17. DIRECTIONS *for questions 17 to 19:* Select the correct alternative from the given choices.

A number has 28 factors. If n of these are perfect squares, n cannot be

☐ a) 4.

☐ b) 6.

☐ c) 7.

☐ d) 8.

Q18. DIRECTIONS *for questions 17 to 19:* Select the correct alternative from the given choices.

If the average of 60 numbers is 40, whereas the average of 40 other numbers is 60, what is the average of all the 100 numbers taken together?

☐ a) 48

☐ b) 50

☐ c) 52

☐ d) 54

Q19. DIRECTIONS *for questions 17 to 19:* Select the correct alternative from the given choices.

If the a^{th} , b^{th} , c^{th} terms of a geometric progression are p , q , r respectively, then $p^{b-c} \cdot q^{c-a} \cdot r^{a-b} =$

- ☐ a) 0.
- ☐ b) 1.
- ☐ c) $(a - b)(b - c)(c - a)$.
- ☐ d) abc

Q20. DIRECTIONS *for question 20:* Type in your answer in the input box provided below the question.
Among all the four-digit natural numbers divisible by 24, in how many numbers does the number 24 appear?

Q21. DIRECTIONS *for question 21:* Select the correct alternative from the given choices.

The cost of an eraser is one-twelfth of that of a pen and two-thirds of that of a pencil. If the cost of two pens is ₹33 more than the cost of five pencils, then find the cost of one pen, two pencils and three erasers.

☐ a) ₹30

☐ b) ₹36

☐ c) ₹48

☐ d) ₹33

Q22. DIRECTIONS *for question 22:* Type in your answer in the input box provided below the question.

If N is the greatest possible number which leaves remainders of 3 and 7 when it divides 171 and 247 respectively, what is the remainder when N divides 800?